

Introduction to Artificial Intelligence

Scenario-Based Examination Questions

15-Week Course (Weeks 1-15)

15 Weeks • 75 Scenario Questions • 20 Marks per Question

Instructions to candidates

1. Each scenario question carries **20 marks**.
2. Read each scenario carefully and answer all of its sub-questions.
3. Marks for each sub-question are shown in brackets. No sub-question carries more than 4 marks.
4. Use examples from the scenario wherever possible.

Weeks

Week 1: Introduction to Artificial Intelligence
Week 2: Python, Data and the Machine Learning Workflow
Week 3: Problem Solving, Agents and Uninformed Search
Week 4: Smart Search — Heuristics, Local Search, Games and Constraints
Week 5: Knowledge Representation, Logic, Reasoning and Planning
Week 6: Uncertainty, Probability and Fuzzy Logic
Week 7: Expert Systems and the Analytic Hierarchy Process
Week 8: Supervised Learning
Week 9: Unsupervised, Semi-Supervised and Self-Supervised Learning
Week 10: Reinforcement Learning
Week 11: Neural Networks and Deep Learning
Week 12: Computer Vision
Week 13: NLP, Transformers and Large Language Models
Week 14: Generative AI, Prompt Engineering, RAG and AI Agents
Week 15: Robotics, IoT, Explainable AI and Ethics

Week 1: Introduction to Artificial Intelligence

Question 1.1: The Kampala Startup Pitch

A Kampala startup is pitching an "AI-powered" farming app to investors. One investor asks what AI actually means, whether the app uses machine learning or deep learning, and whether it is "as smart as a human".

- a) Define Artificial Intelligence and give an everyday example. (4 marks)
- b) Explain the relationship between AI, Machine Learning and Deep Learning, with an example of each. (4 marks)
- c) Distinguish between narrow AI and general AI (AGI). (3 marks)
- d) Explain cognitive computing. (3 marks)
- e) Answer the investor's question about whether the app is "as smart as a human". (2 marks)
- f) State the event in 1956 that gave AI its name. (2 marks)
- g) State the three factors that brought AI back after the AI winters. (2 marks)

[Total: 20 marks]

Question 1.2: The AI Timeline Exhibition

Your department is building a wall display on the history of AI for an open day.

- a) Explain the significance of Alan Turing's 1950 proposal. (3 marks)
- b) Explain why the AI winters of the 1970s-1980s happened. (3 marks)
- c) Explain the significance of IBM's Deep Blue in 1997. (3 marks)
- d) Explain the significance of the 2012 ImageNet result. (3 marks)
- e) Explain the significance of AlphaGo in 2016. (3 marks)
- f) Explain the significance of generative AI in the 2020s. (3 marks)
- g) State what happened at the Dartmouth Conference. (2 marks)

[Total: 20 marks]

Question 1.3: The WhatsApp Chatbot Test

A student sets up a test in which classmates chat on WhatsApp with two hidden contacts for ten minutes: one is a classmate and one is a chatbot. Most testers cannot tell which is which.

- a) Explain the Turing Test. (4 marks)
- b) Evaluate whether passing this test proves the chatbot is intelligent. (4 marks)
- c) Explain why researchers now look for better tests of intelligence. (4 marks)

- d)** Explain whether the chatbot is narrow AI or general AI. **(4 marks)**
- e)** Suggest how the test could be improved. **(4 marks)**

[Total: 20 marks]

Question 1.4: AI in a Student's Week

A student lists five tools she used last week: Google Maps, her phone's predictive keyboard, her TikTok feed, face unlock, and a rule-based USSD menu for checking her mobile money balance.

- a)** Define rule-based AI. **(2 marks)**
- b)** Classify Google Maps and the predictive keyboard as rule-based AI, machine learning or deep learning, and justify each. **(3 marks)**
- c)** Classify the TikTok feed and face unlock, and justify each. **(4 marks)**
- d)** Classify the USSD menu and justify it. **(3 marks)**
- e)** Explain whether every AI system is a machine learning system, using an example. **(4 marks)**
- f)** Explain two applications of AI in Africa. **(4 marks)**

[Total: 20 marks]

Question 1.5: The AI Career Fair

At a university career fair, first-year students meet representatives from Sunbird AI, Safaricom, Google Research Africa and Microsoft's Africa Development Centre.

- a)** Explain how Sunbird AI and Safaricom use AI. **(4 marks)**
- b)** Explain the work of Google Research Africa and Microsoft's Africa Development Centre. **(3 marks)**
- c)** Describe the role of an AI Trainer or Data Annotator. **(3 marks)**
- d)** Explain two entry-level routes into AI work for students. **(4 marks)**
- e)** Explain why students should check a company's current website before citing it. **(2 marks)**
- f)** Propose a plan for a first-year student to start building an AI career. **(4 marks)**

[Total: 20 marks]

Week 2: Python, Data and the Machine Learning Workflow

Question 2.1: Predicting Malaria Drug Stock-Outs

A district health office wants to predict malaria drug stock-outs using monthly data exported from DHIS2. District names appear as "Kampala", "kampala" and "KLA" in different records.

- a) Explain "garbage in, garbage out" in this context. (4 marks)
- b) List and briefly describe the seven stages of a machine learning project. (4 marks)
- c) Explain the data collection stage for this project. (3 marks)
- d) Explain how pre-processing would fix the inconsistent district names. (3 marks)
- e) Define a feature. (2 marks)
- f) Give one example of feature engineering for this project. (2 marks)
- g) Explain what deployment would mean for this project. (2 marks)

[Total: 20 marks]

Question 2.2: Five Kinds of Data

A hospital has a patient spreadsheet, WhatsApp voice notes from community health workers, JSON records from a mobile money API, cassava leaf photos tagged by disease, and 10,000 transactions with no fraud tags.

- a) Explain structured data using the hospital's example. (3 marks)
- b) Explain unstructured data. (3 marks)
- c) Explain semi-structured data. (3 marks)
- d) Explain labelled data. (3 marks)
- e) Explain unlabelled data. (3 marks)
- f) Explain why the type of data affects how a model is built. (3 marks)
- g) Give one example of structured data from your own life. (2 marks)

[Total: 20 marks]

Question 2.3: The Missing Clinic Reports

In a dataset of clinic reports, 12% of rows have missing values. Some clinics never report at all.

- a) Explain deleting rows with missing values and its drawback. (4 marks)
- b) Explain three ways of imputing missing values. (4 marks)

- c)** Explain flagging missing values and why missingness can be informative. **(4 marks)**
- d)** Recommend an approach for the clinic data and justify it. **(4 marks)**
- e)** Explain one risk of each approach. **(4 marks)**

[Total: 20 marks]

Question 2.4: Splitting the Loan Data

A fintech has 5,000 labelled loan records and plans an 80/20 train-test split. A junior analyst wants to use the test set to tune the model's settings.

- a)** Define the test set. **(2 marks)**
- b)** Calculate the number of records in the training and test sets. **(3 marks)**
- c)** Explain the "surprise exam" analogy for the test set. **(4 marks)**
- d)** Explain hyperparameter tuning and cross-validation. **(3 marks)**
- e)** Explain what goes wrong if the test set is used for tuning. **(4 marks)**
- f)** Propose a correct workflow for the analyst. **(4 marks)**

[Total: 20 marks]

Question 2.5: The First Python Lab

Students are given a small CSV file of fruit weight, colour and type, with some missing weights, and must prepare it for a model in Jupyter Notebook.

- a)** Explain the role of Pandas in this lab. **(4 marks)**
- b)** Explain the role of NumPy. **(3 marks)**
- c)** Explain the role of Matplotlib and Seaborn. **(3 marks)**
- d)** Explain the role of Scikit-learn, including `train_test_split`. **(4 marks)**
- e)** Explain why Jupyter Notebook is useful for learning. **(2 marks)**
- f)** Write the steps the students should follow to complete the lab. **(4 marks)**

[Total: 20 marks]

Week 3: Problem Solving, Agents and Uninformed Search

Question 3.1: Makerere to Entebbe

A student wants a program that finds a route from Makerere to Entebbe Airport through Kampala's road junctions.

- a) Explain the initial state, goal state, actions and state space for this problem. **(4 marks)**
- b) Explain what an agent is and how the route-finding program acts as one. **(4 marks)**
- c) Explain path cost with examples for this trip. **(3 marks)**
- d) Explain uninformed search using the "looking for your keys" analogy. **(3 marks)**
- e) Define an algorithm. **(2 marks)**
- f) Give an example of a state in this problem. **(2 marks)**
- g) Identify the initial state, goal state and actions for making a cup of tea. **(2 marks)**

[Total: 20 marks]

Question 3.2: The Road to Jinja

A map has these roads: Kampala-Mukono 20 km, Kampala-Wakiso 15 km, Mukono-Jinja 60 km, Wakiso-Luwero 40 km and Luwero-Jinja 20 km. You must travel from Kampala to Jinja.

- a) Explain how Breadth-First Search works. **(3 marks)**
- b) State the route BFS finds and explain why. **(3 marks)**
- c) Explain how Uniform Cost Search works. **(3 marks)**
- d) State the route UCS finds and its total distance. **(3 marks)**
- e) Compare the distances of the BFS and UCS routes. **(3 marks)**
- f) Explain how Depth-First Search would behave on this map. **(3 marks)**
- g) Explain Bidirectional Search. **(2 marks)**

[Total: 20 marks]

Question 3.3: Choosing a Search Method

A developer faces four situations: a device with very little memory; a traveller who wants the fewest bus changes; a traveller who wants the cheapest total fare; and a very large national road map with a clear destination.

- a) Explain the strengths and weaknesses of BFS. **(4 marks)**

- b)** Explain the strengths and weaknesses of DFS. **(4 marks)**
- c)** Explain the strengths and weaknesses of UCS. **(4 marks)**
- d)** Explain the strengths and weaknesses of Bidirectional Search. **(4 marks)**
- e)** Match each situation to the most suitable method and justify each. **(4 marks)**

[Total: 20 marks]

Question 3.4: The Delivery Rider

A SafeBoda delivery rider must visit several customers around Kampala. The company wants software to plan the rider's route.

- a)** Define an agent. **(2 marks)**
- b)** Describe the states and actions for the rider's problem. **(3 marks)**
- c)** Explain three ways the path cost could be measured. **(4 marks)**
- d)** Explain why uninformed search is slow on a large city map. **(3 marks)**
- e)** Explain how shortest-path search is used in network routing. **(4 marks)**
- f)** Describe two companies and two jobs that use search algorithms. **(4 marks)**

[Total: 20 marks]

Question 3.5: The 8-Queens Puzzle

A student must place 8 queens on a chessboard so that no two queens attack each other.

- a)** Describe the state space for this puzzle. **(4 marks)**
- b)** Explain how DFS with backtracking would solve it. **(3 marks)**
- c)** Explain why BFS would use a lot of memory here. **(3 marks)**
- d)** Explain the rule that limits where queens can go. **(4 marks)**
- e)** Define the goal state. **(2 marks)**
- f)** Describe how you would solve a smaller 4-queens version on paper. **(4 marks)**

[Total: 20 marks]

Week 4: Smart Search — Heuristics, Local Search, Games and Constraints

Question 4.1: Choosing the Next Node

During a route search, two nodes are waiting to be expanded. Node N has $g = 5$ and $h = 10$. Node M has $g = 14$ and $h = 4$.

- a) Explain what a heuristic is, using the kitchen-smell example. (4 marks)
- b) Calculate $f(n) = g(n) + h(n)$ for both nodes. (4 marks)
- c) State which node A* expands next and why. (3 marks)
- d) State which node Greedy Best-First Search expands next and why. (3 marks)
- e) State the condition for A* to find the best path. (2 marks)
- f) Explain why straight-line distance is a safe heuristic. (2 marks)
- g) Explain why A* is better than greedy best-first search. (2 marks)

[Total: 20 marks]

Question 4.2: Arranging Owino Market

Market managers want to arrange stalls in Owino market to reduce the distance shoppers walk. They start with any layout and swap stalls to improve it.

- a) Explain hill climbing. (3 marks)
- b) Explain how hill climbing can get stuck at a local maximum. (3 marks)
- c) Explain simulated annealing. (3 marks)
- d) Explain the metal-cooling idea behind simulated annealing. (3 marks)
- e) Explain local beam search. (3 marks)
- f) Recommend a local search method for the market and justify it. (3 marks)
- g) Define a neighbouring solution. (2 marks)

[Total: 20 marks]

Question 4.3: Game-Playing AI

A draughts AI has two moves. Move A leads to opponent replies worth 3 and 7 to the AI. Move B leads to opponent replies worth 5 and 6. The AI is the maximising player.

- a) Explain the minimax idea. (4 marks)
- b) Calculate the minimax value of each move and state the AI's choice. (4 marks)
- c) Explain alpha-beta pruning. (4 marks)
- d) If Move B is evaluated first, identify which leaf can be pruned and explain why. (4 marks)

- e) Explain how minimax could be applied to Omweso. (4 marks)

[Total: 20 marks]

Question 4.4: The Exam Timetable

A university must schedule 40 exams into available time slots. No student may sit two exams at once, and the main hall holds 200 people.

- a) Define a Constraint Satisfaction Problem. (2 marks)
- b) Identify the variables, values and constraints in this timetable. (3 marks)
- c) Explain backtracking search. (4 marks)
- d) Explain constraint propagation. (3 marks)
- e) Explain the Sudoku pencil-mark analogy for constraint propagation. (4 marks)
- f) Design the variables, values and constraints for a hospital nurse roster. (4 marks)

[Total: 20 marks]

Question 4.5: Vaccines for Rural Tanzania

A health programme must deliver vaccines to rural Tanzanian clinics. Vaccines need cold storage, and some roads become impassable in the rainy season.

- a) Describe the states, actions and costs for this problem. (4 marks)
- b) Explain the constraints the system must respect. (3 marks)
- c) Recommend a search algorithm and justify it. (3 marks)
- d) Explain how Zipline plans drone deliveries of medical supplies. (4 marks)
- e) State what UPS's ORION system does. (2 marks)
- f) Design a search system for the vaccine programme. (4 marks)

[Total: 20 marks]

Week 5: Knowledge Representation, Logic, Reasoning and Planning

Question 5.1: The Fees Rule

A university system stores the rule "All students must pay fees before exams" and the fact "Amina is a student". It must conclude what applies to Amina.

- a) Explain knowledge representation. (4 marks)
- b) Write a propositional logic rule for "if it is raining and I have no umbrella, I get wet", and explain the connectives. (4 marks)
- c) Write the fees rule in first-order logic. (3 marks)
- d) Explain the difference between propositional logic and first-order logic. (3 marks)
- e) Define inference. (2 marks)
- f) Define an inference engine. (2 marks)
- g) State the conclusion the system draws about Amina. (2 marks)

[Total: 20 marks]

Question 5.2: Three Kinds of Reasoning

A lecturer gives three statements: "All mammals breathe air; a whale is a mammal; so a whale breathes air." "The sun has risen every morning for a year; so it rises every morning." "The grass is wet; so it probably rained."

- a) Explain deductive reasoning using the first statement. (3 marks)
- b) Explain inductive reasoning using the second statement. (3 marks)
- c) Explain abductive reasoning using the third statement. (3 marks)
- d) Classify a doctor diagnosing malaria from symptoms and justify it. (3 marks)
- e) Explain why inductive conclusions can be wrong. (3 marks)
- f) Identify which type of reasoning gives certain conclusions and explain why. (3 marks)
- g) Give your own example of abductive reasoning. (2 marks)

[Total: 20 marks]

Question 5.3: Digitising Land Records

Uganda's land offices want to digitise land records to reduce ownership disputes. Records link parcels, owners, titles and transfers, but some older records are incomplete.

- a) Define an ontology. (4 marks)

- b)** Describe the main concepts and relationships the land ontology should contain. **(4 marks)**
- c)** Explain how the ontology could reduce land disputes. **(4 marks)**
- d)** Compare a semantic network with an ontology. **(4 marks)**
- e)** Explain how the system should behave when a fact is missing or uncertain. **(4 marks)**

[Total: 20 marks]

Question 5.4: Frames and Knowledge Graphs

A university IT team wants to store knowledge about patients at its clinic, farmers in its outreach programme, and its own departments, lecturers and courses.

- a)** Define a frame. **(2 marks)**
- b)** Describe a "Patient" frame and its slots. **(3 marks)**
- c)** Design a "Farmer" frame with at least five slots. **(4 marks)**
- d)** Explain how a knowledge graph could represent the university. **(3 marks)**
- e)** Explain how the graph could answer "Which courses does Dr X teach?" **(4 marks)**
- f)** Explain how Google's Knowledge Graph and Africa Check use structured knowledge. **(4 marks)**

[Total: 20 marks]

Question 5.5: The Warehouse Robot's Plan

A warehouse robot has the goal "box A on the truck". It starts at its charging point.

- a)** Explain AI planning. **(4 marks)**
- b)** Explain forward state-space search. **(3 marks)**
- c)** Write a plan for the robot to achieve its goal. **(3 marks)**
- d)** Explain how planning relates to the search methods of Week 3. **(4 marks)**
- e)** Describe the role of a Knowledge Engineer. **(2 marks)**
- f)** Design a plan for a robot delivering medicines to three wards in a hospital. **(4 marks)**

[Total: 20 marks]

Week 6: Uncertainty, Probability and Fuzzy Logic

Question 6.1: Hot and Humid

On a given afternoon, the degree to which the weather is "hot" is 0.6 and the degree to which it is "humid" is 0.9.

- a) Explain the difference between probability and fuzzy membership. (4 marks)
- b) Compare crisp and fuzzy definitions of "tall" using heights of 190 cm and 178 cm. (4 marks)
- c) Calculate "hot AND humid" and "hot OR humid". (3 marks)
- d) Calculate "NOT hot" and "NOT humid". (3 marks)
- e) Define a membership function. (2 marks)
- f) Name two common shapes of membership function. (2 marks)
- g) Give an example of probabilistic reasoning in health care. (2 marks)

[Total: 20 marks]

Question 6.2: The Smart Fan

A fuzzy fan controller reads a temperature of 32 °C, which it treats as "warm" to degree 0.2 and "hot" to degree 0.8. It outputs a fan speed in rpm.

- a) Explain fuzzification using this reading. (3 marks)
- b) Explain rule evaluation. (3 marks)
- c) Explain aggregation. (3 marks)
- d) Explain defuzzification. (3 marks)
- e) Explain the Mamdani model. (3 marks)
- f) Explain the Sugeno model. (3 marks)
- g) Recommend a model for fast real-time control. (2 marks)

[Total: 20 marks]

Question 6.3: A Busy Kampala Junction

A fuzzy traffic-light controller uses queue length (short, medium, long) and waiting time (low, high). At one moment, "queue is long" = 0.7 and "waiting time is high" = 0.5.

- a) Describe the inputs and their fuzzy sets. (4 marks)
- b) Write three IF-THEN rules for the controller. (4 marks)
- c) Calculate the strength of the rule "IF queue is long AND waiting time is high THEN green is long". (4 marks)
- d) Explain how the controller turns this into a green-light time. (4 marks)

- e) Compare the fuzzy controller with a fixed-timer traffic light. (4 marks)

[Total: 20 marks]

Question 6.4: Smart Irrigation

A solar irrigation company wants its pumps to follow rules like "IF soil is fairly dry AND forecast is little rain THEN water moderately".

- a) Distinguish between a crisp rule and a fuzzy rule. (2 marks)
- b) Identify the inputs and output of the system. (3 marks)
- c) Write three fuzzy rules for the system. (4 marks)
- d) Explain why fuzzy logic suits the way farmers describe conditions. (3 marks)
- e) Explain two other applications of fuzzy logic in everyday devices. (4 marks)
- f) Describe two jobs that use fuzzy logic skills. (4 marks)

[Total: 20 marks]

Question 6.5: Triage in a Rural Clinic

A rural clinic wants a system to help health workers prioritise patients who describe symptoms vaguely, such as "mild fever" or "severe pain".

- a) Explain how fuzzy logic could support triage. (4 marks)
- b) Explain the risks of using such a system. (3 marks)
- c) Explain how probabilistic reasoning could be used instead. (3 marks)
- d) Explain how probability and fuzzy logic could be combined. (4 marks)
- e) Explain what "degree of truth" means. (2 marks)
- f) Design a simple fuzzy triage system with inputs, rules and output. (4 marks)

[Total: 20 marks]

Week 7: Expert Systems and the Analytic Hierarchy Process

Question 7.1: The Tomato Disease Advisor

An agricultural extension office wants a phone-based expert system that asks farmers questions about their tomato plants and suggests the likely disease.

- a) Define an expert system. (4 marks)
- b) Explain the four components of an expert system. (4 marks)
- c) Explain forward chaining using the tomato example. (3 marks)
- d) Explain backward chaining using the tomato example. (3 marks)
- e) State what MYCIN did. (2 marks)
- f) State what XCON did. (2 marks)
- g) Explain why the explanation facility matters to farmers. (2 marks)

[Total: 20 marks]

Question 7.2: Rules or Learning?

A SACCO must decide whether to automate loan approval with an expert system built from its loan officers' rules or with a machine learning model trained on past loans. It has records for only 300 past loans.

- a) Compare where knowledge comes from in each approach. (3 marks)
- b) Compare their data needs. (3 marks)
- c) Compare their explainability. (3 marks)
- d) Compare how they handle new situations. (3 marks)
- e) Explain when an expert system is the better choice. (3 marks)
- f) Explain when machine learning is the better choice. (3 marks)
- g) Recommend an approach for the SACCO. (2 marks)

[Total: 20 marks]

Question 7.3: Choosing a Hostel with AHP

A student uses AHP to choose a hostel. Her weights are security 55%, price 30% and distance 15%. Scores out of 10 are: Hostel A — security 8, price 5, distance 6; Hostel B — 6, 8, 7; Hostel C — 9, 4, 3.

- a) Describe the AHP hierarchy for this decision. (4 marks)
- b) Explain how pairwise comparisons on the 1-9 scale produce the weights. (4 marks)

- c)** Calculate the overall score of each hostel and rank them. **(4 marks)**
- d)** Explain the consistency check. **(4 marks)**
- e)** Recalculate the scores if the weights change to security 45%, price 40% and distance 15%, and comment on the result. **(4 marks)**

[Total: 20 marks]

Question 7.4: Inconsistent Judgements

A planning committee judges that site A is more important than site B, B more than C, but C more than A. Its consistency ratio comes out at 0.18.

- a)** Define consistency in AHP. **(2 marks)**
- b)** Explain why the committee's judgements are inconsistent. **(3 marks)**
- c)** Explain the consistency ratio threshold. **(4 marks)**
- d)** Explain what the committee should do next. **(3 marks)**
- e)** Explain the 1–9 comparison scale. **(4 marks)**
- f)** Propose criteria and a hierarchy for choosing a site for a new health centre. **(4 marks)**

[Total: 20 marks]

Question 7.5: Expert Systems at Work

A report reviews PlantVillage Nuru, digital clinical decision support tools used by ministries of health, and AHP studies used in Rwanda's urban planning.

- a)** Explain how PlantVillage Nuru supports farmers. **(4 marks)**
- b)** Explain how rule-based clinical decision support helps health workers. **(3 marks)**
- c)** Explain how AHP helps prioritise urban projects. **(3 marks)**
- d)** Describe two jobs that use expert systems or AHP. **(4 marks)**
- e)** Explain how frames organise knowledge in an expert system. **(2 marks)**
- f)** Plan how you would build and test an eight-rule tomato disease system. **(4 marks)**

[Total: 20 marks]

Week 8: Supervised Learning

Question 8.1: Scoring Mobile Loans

A mobile lender has 5,000 past loan applications, each labelled "repaid" or "defaulted", with features such as income, M-Pesa activity and past loans.

- a) Explain supervised learning using the teacher analogy. (4 marks)
- b) Distinguish between classification and regression. (4 marks)
- c) Classify "predict tomorrow's rainfall in millimetres" and justify it. (3 marks)
- d) Classify "predict whether an applicant will default" and justify it. (3 marks)
- e) Define a label. (2 marks)
- f) Give two features the lender could use. (2 marks)
- g) Explain the role of the 5,000 past applications. (2 marks)

[Total: 20 marks]

Question 8.2: Which Algorithm?

A data science team is matching algorithms to tasks: predicting maize yield, spam detection, explaining decisions to loan officers, accurate predictions on table data, text classification with small data, simple recommendations, and winning a Zindi competition.

- a) Explain linear regression and match it to a task. (3 marks)
- b) Explain logistic regression and match it to a task. (3 marks)
- c) Explain decision trees and match them to a task. (3 marks)
- d) Explain random forests and match them to a task. (3 marks)
- e) Explain SVMs and match them to a task. (3 marks)
- f) Explain k-NN and Naive Bayes and match each to a task. (3 marks)
- g) Explain gradient boosting and match it to a task. (2 marks)

[Total: 20 marks]

Question 8.3: Catching Mobile Money Fraud

A fraud model is tested on 10,000 transactions, 100 of which are fraud. It catches 70 fraud cases, misses 30, and wrongly flags 130 genuine transactions.

- a) Calculate the accuracy. (4 marks)
- b) Calculate the precision. (4 marks)
- c) Calculate the recall. (4 marks)
- d) Calculate the F1-score. (4 marks)

- e) Calculate the accuracy and recall of a model that always predicts "not fraud", and explain why accuracy is misleading. **(4 marks)**

[Total: 20 marks]

Question 8.4: The Overfitted Tree

A decision tree scores 98% on training data but only 70% on test data.

- a) Define overfitting. **(2 marks)**
- b) Explain overfitting using the past-paper analogy. **(3 marks)**
- c) Explain underfitting and how it differs. **(4 marks)**
- d) Diagnose the tree's problem from its scores. **(3 marks)**
- e) Explain four ways to fix overfitting. **(4 marks)**
- f) Design a lab experiment to show overfitting as tree depth increases. **(4 marks)**

[Total: 20 marks]

Question 8.5: Predicting House Rent

A rent-prediction model is tested on two houses with actual rents of 400 and 600 (thousand UGX). It predicts 450 and 550.

- a) Calculate the Mean Squared Error. **(4 marks)**
- b) Calculate the Root Mean Squared Error. **(3 marks)**
- c) Explain what R-squared measures. **(3 marks)**
- d) Recommend which metric to report to a landlord and justify it. **(4 marks)**
- e) State what Zillow's "Zestimate" does. **(2 marks)**
- f) Describe two jobs that build and evaluate supervised models. **(4 marks)**

[Total: 20 marks]

Week 9: Unsupervised, Semi-Supervised and Self-Supervised Learning

Question 9.1: Five Million Customers

A telecom has 5 million customers and no labels. It wants to discover customer groups to design new bundles.

- a) Explain unsupervised learning. (4 marks)
- b) Explain clustering using the fruit basket analogy. (4 marks)
- c) Explain how K-Means clustering works. (3 marks)
- d) Explain how the elbow method helps choose K. (3 marks)
- e) Explain hierarchical clustering. (2 marks)
- f) Explain DBSCAN. (2 marks)
- g) Give two customer groups the telecom might discover. (2 marks)

[Total: 20 marks]

Question 9.2: Matching the Method to the Task

A research team has five tasks: grouping districts by health risk; building a family tree of crop varieties; spotting unusual transactions; simplifying 50 soil measurements; and detecting a broken sensor.

- a) Explain K-Means and match it to a task. (3 marks)
- b) Explain hierarchical clustering and match it to a task. (3 marks)
- c) Explain DBSCAN and match it to a task. (3 marks)
- d) Explain PCA and match it to a task. (3 marks)
- e) Explain anomaly detection and match it to a task. (3 marks)
- f) Explain the "one-page report" analogy for PCA. (3 marks)
- g) Define an outlier. (2 marks)

[Total: 20 marks]

Question 9.3: The X-Ray Archive

A hospital has 200 chest X-rays labelled by a radiologist and 20,000 unlabelled X-rays.

- a) Define semi-supervised learning. (4 marks)
- b) Explain the "learning words from context" analogy. (4 marks)
- c) Explain how semi-supervised learning saves the hospital money. (4 marks)
- d) Explain two risks of this approach. (4 marks)

- e) Compare this approach with training only on the 200 labelled X-rays. (4 marks)

[Total: 20 marks]

Question 9.4: A Luganda Language Model

A startup has millions of Luganda radio transcripts and audio recordings but very few labelled examples.

- a) Define self-supervised learning. (2 marks)
- b) Explain how hiding a word in a sentence creates a training task. (3 marks)
- c) Explain how self-supervised learning is used to build large language models. (4 marks)
- d) Explain why self-supervised learning matters for low-resource African languages. (3 marks)
- e) Explain two sources of raw text or audio the startup could use. (4 marks)
- f) Propose a plan to build a useful Luganda model. (4 marks)

[Total: 20 marks]

Question 9.5: Patterns in Business and Health

A consultancy is presenting unsupervised learning to a telecom, a card network, a music platform and a health ministry.

- a) Explain how customer segmentation helps a telecom. (4 marks)
- b) Explain how anomaly detection helps card networks such as Visa and Mastercard. (3 marks)
- c) Explain how Spotify uses clustering. (3 marks)
- d) Explain how clustering helps an epidemiology team. (4 marks)
- e) Describe the role of a CRM Analyst. (2 marks)
- f) Design an unsupervised learning project for the health ministry. (4 marks)

[Total: 20 marks]

Week 10: Reinforcement Learning

Question 10.1: Lagos Traffic Lights

A city wants a traffic-light controller that learns timings by trying them in a simulator, earning rewards for shorter queues.

- a) Explain reinforcement learning. (4 marks)
- b) Identify the agent, environment, state, action and reward in this system. (4 marks)
- c) Explain what the controller's policy is. (3 marks)
- d) Explain how the reward should be designed. (3 marks)
- e) Explain the dog-training analogy for RL. (2 marks)
- f) Explain how RL differs from supervised learning. (2 marks)
- g) Explain what a delayed reward is. (2 marks)

[Total: 20 marks]

Question 10.2: Ambulance Dispatch

A rural district wants an RL system to choose which ambulance to send and which route to take.

- a) Identify the agent. (3 marks)
- b) Describe the environment. (3 marks)
- c) Describe the state. (3 marks)
- d) Describe the actions. (3 marks)
- e) Design the reward. (3 marks)
- f) Explain exploration and exploitation in this system. (3 marks)
- g) Explain the policy the system learns. (2 marks)

[Total: 20 marks]

Question 10.3: Thinking Ahead

An agent receives a reward of 5 now and moves to a state whose value is 20. The discount factor is 0.9.

- a) Explain a Markov Decision Process. (4 marks)
- b) Explain the Markov property. (4 marks)
- c) Use the Bellman equation to calculate the value of the current state. (4 marks)
- d) Explain how Q-learning uses a table of Q-values. (4 marks)
- e) Explain why Deep Q-Networks are needed for video games. (4 marks)

[Total: 20 marks]

Question 10.4: Reward Hacking

A traffic RL agent is rewarded only for short queues on the main road. After training, side-road drivers and pedestrians complain.

- a)** Define reward hacking. **(2 marks)**
- b)** Explain what has happened to side roads and pedestrians. **(3 marks)**
- c)** Propose a better reward. **(4 marks)**
- d)** Explain why real-world trial and error can be unsafe. **(3 marks)**
- e)** Discuss the ethical responsibilities of the designers. **(4 marks)**
- f)** Propose a safe plan to train and deploy the controller. **(4 marks)**

[Total: 20 marks]

Question 10.5: RL in Industry

A lecture reviews Google DeepMind's data-centre cooling, Ocado's warehouse robots, chatbot fine-tuning with human feedback, and InstaDeep's logistics work.

- a)** Explain how RL helps cool data centres. **(4 marks)**
- b)** Explain how Ocado uses RL-style coordination in warehouses. **(3 marks)**
- c)** Explain reinforcement learning from human feedback (RLHF). **(3 marks)**
- d)** Explain how InstaDeep applies RL. **(4 marks)**
- e)** Name one Gymnasium environment used to practise RL. **(2 marks)**
- f)** Propose an RL project that could benefit Uganda. **(4 marks)**

[Total: 20 marks]

Week 11: Neural Networks and Deep Learning

Question 11.1: A Single Neuron

A neuron receives inputs $x_1 = 2$ and $x_2 = 3$ with weights $w_1 = 0.5$ and $w_2 = -0.2$, and a bias of 0.1.

- a) Explain what a neural network is, using the relay-race analogy. (4 marks)
- b) Calculate the neuron's weighted sum. (4 marks)
- c) Calculate the output after a ReLU activation. (3 marks)
- d) Recalculate the ReLU output if the bias changes to -0.7 , and explain the result. (3 marks)
- e) Define weights. (2 marks)
- f) Define an epoch. (2 marks)
- g) Explain what makes a network "deep". (2 marks)

[Total: 20 marks]

Question 11.2: How the Network Learns

A student's network is making poor predictions. The lecturer explains how training fixes this.

- a) Explain the loss function. (3 marks)
- b) Explain backpropagation using the coach analogy. (3 marks)
- c) Explain gradient descent. (3 marks)
- d) Compare the ReLU and sigmoid activation functions. (3 marks)
- e) Explain the vanishing gradient problem. (3 marks)
- f) Explain why deep learning needs large datasets and is prone to overfitting. (3 marks)
- g) Explain why GPUs are used for deep learning. (2 marks)

[Total: 20 marks]

Question 11.3: Choosing an Architecture

A team has five projects: detecting cassava disease from leaf photos; forecasting rainfall from months of weather data; predicting electricity demand on a low-power device; predicting the next word a user types; and translating English to Swahili.

- a) Explain CNNs and match them to a project. (4 marks)
- b) Explain LSTMs and match them to a project. (4 marks)
- c) Explain GRUs and match them to a project. (4 marks)

- d)** Explain RNNs and match them to a project. **(4 marks)**
- e)** Explain transformers and match them to a project. **(4 marks)**

[Total: 20 marks]

Question 11.4: Only 400 Local Images

A lab has only 400 labelled photos of a local crop disease, not enough to train a deep network from scratch.

- a)** Define transfer learning. **(2 marks)**
- b)** Explain why transfer learning helps when data is scarce. **(3 marks)**
- c)** Describe the steps the lab should follow. **(4 marks)**
- d)** Explain the risk if the pre-trained model's images are very different from local conditions. **(3 marks)**
- e)** Explain how an Edge AI Engineer would prepare the model to run on cheap phones. **(4 marks)**
- f)** Propose a plan for building and testing the model. **(4 marks)**

[Total: 20 marks]

Question 11.5: Recognising Handwritten Digits

A student designs a network for MNIST digits with 784 input pixels, one hidden layer of 128 neurons, and 10 output neurons.

- a)** Explain the role of each layer and why it has that size. **(4 marks)**
- b)** Calculate the number of weights between the input and hidden layers. **(3 marks)**
- c)** Calculate the number of weights between the hidden and output layers, and the total number of weights and biases. **(3 marks)**
- d)** Design an experiment comparing ReLU and sigmoid. **(4 marks)**
- e)** Explain what the 10 output neurons represent. **(2 marks)**
- f)** Explain how the student should evaluate the trained network. **(4 marks)**

[Total: 20 marks]

Week 12: Computer Vision

Question 12.1: How Machines See

A farmer asks how a phone app can "see" that a cassava leaf is diseased.

- a) Define computer vision. (4 marks)
- b) Explain how a computer stores an image as numbers. (4 marks)
- c) Distinguish between image classification and object detection. (3 marks)
- d) Explain image segmentation. (3 marks)
- e) Explain OCR. (2 marks)
- f) Explain face recognition. (2 marks)
- g) Explain feature extraction. (2 marks)

[Total: 20 marks]

Question 12.2: Six Vision Projects

A vision company has six projects: diagnosing cassava brown streak disease; counting boda-bodas at a junction; mapping flooded land from satellite images; reading number plates; verifying national ID cards for bank sign-up; and finding edges in a factory inspection image.

- a) Identify the task for the cassava project and justify it. (3 marks)
- b) Identify the task for the boda-boda project and justify it. (3 marks)
- c) Identify the task for the flood project and justify it. (3 marks)
- d) Identify the task for the number-plate project and justify it. (3 marks)
- e) Identify the tasks for the ID project and justify them. (3 marks)
- f) Identify the task for the factory project and justify it. (3 marks)
- g) State how Smile ID uses computer vision. (2 marks)

[Total: 20 marks]

Question 12.3: Real-Time Detection with YOLO

A road-safety team wants to count people and vehicles in live video of a busy taxi park.

- a) Explain how YOLO works. (4 marks)
- b) Explain why YOLO suits real-time video. (4 marks)
- c) Explain the role of the grid cells. (4 marks)
- d) Explain the taxi-park analogy for YOLO. (4 marks)
- e) Describe how the team could count people and vehicles in a photo. (4 marks)

[Total: 20 marks]

Question 12.4: Tesla on Ugandan Roads

A student asks whether Tesla's camera-based driving system would work on a Ugandan road with boda-bodas, livestock, dust and unmarked lanes.

- a) Describe how Tesla's system perceives the road. (2 marks)
- b) Explain the role of fleet data and edge cases. (3 marks)
- c) Explain why the system still requires an attentive driver. (4 marks)
- d) Compare Tesla's approach with Waymo's. (3 marks)
- e) Explain four challenges the system would face on Ugandan roads. (4 marks)
- f) Explain three lessons students should take from the case study. (4 marks)

[Total: 20 marks]

Question 12.5: Watching for Poachers

A Kenyan wildlife park wants camera traps and drones to spot animals and poachers, day and night.

- a) Explain which computer vision tasks the system needs. (4 marks)
- b) Explain three challenges the system will face. (3 marks)
- c) Explain how training data would be created. (3 marks)
- d) Compare running the model on the camera with running it in the cloud. (4 marks)
- e) Describe the role of an Image Annotation Specialist. (2 marks)
- f) Design the anti-poaching vision system. (4 marks)

[Total: 20 marks]

Week 13: NLP, Transformers and Large Language Models

Question 13.1: "What's the Weather in Gulu?"

A user asks a phone assistant, "Hey Google, what's the weather in Gulu?" and hears a spoken forecast.

- a) Define Natural Language Processing. (4 marks)
- b) Describe the steps the assistant takes to answer. (4 marks)
- c) Explain tokenisation using "Webale nnyo". (3 marks)
- d) Explain part-of-speech tagging. (3 marks)
- e) Explain embeddings. (2 marks)
- f) Explain sentiment analysis. (2 marks)
- g) Explain speech recognition. (2 marks)

[Total: 20 marks]

Question 13.2: The Sick Cow

In the sentence "The farmer sold the cow because it was sick", a model must work out what "it" refers to.

- a) Explain why older models struggled with long sentences. (3 marks)
- b) Explain attention using this sentence. (3 marks)
- c) Explain self-attention. (3 marks)
- d) Explain multi-head attention. (3 marks)
- e) Explain positional encoding. (3 marks)
- f) Distinguish between an encoder and a decoder. (3 marks)
- g) State when and by whom transformers were introduced. (2 marks)

[Total: 20 marks]

Question 13.3: BERT or GPT?

A bank wants to classify incoming SMS messages as scam or genuine, and separately wants to generate summaries of customer complaints.

- a) Explain BERT. (4 marks)
- b) Explain GPT-style models. (4 marks)
- c) Recommend a model family for scam classification and justify it. (4 marks)
- d) Recommend a model family for writing summaries and justify it. (4 marks)

- e) Explain how LLMs are trained and then refined. (4 marks)

[Total: 20 marks]

Question 13.4: When the Chatbot Gets It Wrong

A chatbot confidently gives a wrong answer about a Ugandan law, and performs poorly when users mix English and Luganda.

- a) Define hallucination. (2 marks)
- b) Explain how bias enters LLMs. (3 marks)
- c) Explain why LLMs perform worse in many African languages. (4 marks)
- d) Explain how projects like Masakhane and Mozilla Common Voice help. (3 marks)
- e) Explain code-switching and why it is hard for models. (4 marks)
- f) Propose a plan to test the chatbot before public release. (4 marks)

[Total: 20 marks]

Question 13.5: A Farming Chatbot in Luganda

An agritech startup wants a chatbot that gives farming advice in Luganda by text and voice.

- a) Explain how machine translation supports the chatbot. (4 marks)
- b) Explain how speech recognition supports it. (3 marks)
- c) Explain how the chatbot holds a conversation. (3 marks)
- d) Explain how organisations such as Sunbird AI and Lelapa AI support African language AI. (4 marks)
- e) Describe the role of a Computational Linguist. (2 marks)
- f) Design the chatbot, including its data and testing. (4 marks)

[Total: 20 marks]

Week 14: Generative AI, Prompt Engineering, RAG and AI Agents

Question 14.1: Creating New Content

A teacher asks a chatbot to write a story for Primary 4 pupils about a clever hare in Lira, and uses an image model to illustrate it.

- a) Define generative AI. (4 marks)
- b) Explain how an LLM generates the story. (4 marks)
- c) Explain GANs using the counterfeiter analogy. (3 marks)
- d) Explain how diffusion models create images. (3 marks)
- e) Name one diffusion-based image generator. (2 marks)
- f) Distinguish between a GAN and a diffusion model. (2 marks)
- g) Explain the difference between resembling and copying training data. (2 marks)

[Total: 20 marks]

Question 14.2: Prompting for Scam Detection

A student wants an LLM to classify SMS messages as scam or not scam, and to explain its reasoning for a report to smallholder farmers.

- a) Write a zero-shot prompt for the task. (3 marks)
- b) Explain how a few-shot prompt would differ. (3 marks)
- c) Explain chain-of-thought prompting. (3 marks)
- d) Explain role and context prompting with an example. (3 marks)
- e) Explain format instructions with an example. (3 marks)
- f) Explain three tips for writing good prompts. (3 marks)
- g) Explain why the output must still be checked. (2 marks)

[Total: 20 marks]

Question 14.3: The Malaria Guidelines Chatbot

A Ministry of Health wants a chatbot that answers health workers' questions using the national malaria treatment guidelines.

- a) Explain why a plain LLM might give wrong answers here. (4 marks)
- b) Explain the retrieval step of RAG. (4 marks)
- c) Explain the generation step of RAG. (4 marks)
- d) Explain why answers should cite the guideline page. (4 marks)

e) Explain two limitations of RAG.

(4 marks)

[Total: 20 marks]

Question 14.4: The Clinic Supply Agent

A district wants an AI system that checks which health centres will run out of antimalarials in two weeks and drafts restock orders for a human to approve.

a) Define an AI agent.

(2 marks)

b) Explain the four parts of PEAS.

(3 marks)

c) Describe the clinic-supply agent using PEAS.

(4 marks)

d) Distinguish between reflex and deliberative agents.

(3 marks)

e) Explain tool calling and agent memory in this system.

(4 marks)

f) Explain why a human should approve the orders.

(4 marks)

[Total: 20 marks]

Question 14.5: Deepfakes and Elections

Ahead of an election, fake audio clips of candidates are circulating on WhatsApp.

a) Explain how generative AI can spread election misinformation.

(4 marks)

b) Explain how deepfake detection tools work.

(3 marks)

c) Describe the role of a Trust and Safety or Misinformation Analyst.

(3 marks)

d) Explain four ways to protect election integrity.

(4 marks)

e) Define a multi-agent system.

(2 marks)

f) Propose a response plan for the electoral commission.

(4 marks)

[Total: 20 marks]

Week 15: Robotics, IoT, Explainable AI and Ethics

Question 15.1: The Smart Farm in Zambia

A Zambian farm has soil sensors, an AI model that reads sensor data and weather forecasts to switch a solar pump on only when needed, and a drone that sprays only sick patches.

- a) Explain robotics, sensors and actuators using the farm. (4 marks)
- b) Distinguish between IoT and AIoT using the farm. (4 marks)
- c) Compare edge computing and cloud computing for the farm. (3 marks)
- d) Explain IoRT. (3 marks)
- e) Explain IIoT. (2 marks)
- f) Define cloud computing. (2 marks)
- g) State what AirQo does. (2 marks)

[Total: 20 marks]

Question 15.2: The Rejected Loan

A bank's AI rejects a loan application. SHAP shows the main reasons were "only 2 months of mobile money history" and "high existing debt".

- a) Define explainable AI. (3 marks)
- b) Explain feature importance. (3 marks)
- c) Explain LIME. (3 marks)
- d) Explain SHAP. (3 marks)
- e) Explain why explainability matters more for loans than for film recommendations. (3 marks)
- f) Explain what the applicant can do with the explanation. (3 marks)
- g) Name one African data protection law. (2 marks)

[Total: 20 marks]

Question 15.3: Ethics in Practice

A review finds that an ID system's face recognition works worse on darker skin, a clinic AI tool gave a wrong diagnosis, health records were shared without consent, and a factory automated jobs without retraining workers.

- a) Explain bias and fairness using the face recognition case. (4 marks)
- b) Explain privacy and the relevant data protection laws. (4 marks)
- c) Explain accountability using the wrong diagnosis. (4 marks)

- d)** Explain jobs and inclusion using the factory case. **(4 marks)**
- e)** Explain safety risks of autonomous drones in crowded areas. **(4 marks)**

[Total: 20 marks]

Question 15.4: Robots at the Port

A port such as Mombasa uses networked robots to move containers, and a nearby mine uses vibration sensors on its pumps.

- a)** Define IoRT. **(2 marks)**
- b)** Explain how IoRT helps the port's robots coordinate. **(3 marks)**
- c)** Explain how IIoT supports predictive maintenance at the mine. **(4 marks)**
- d)** Explain the role of ROS in robotics. **(3 marks)**
- e)** Explain how two companies use robotics or IoT. **(4 marks)**
- f)** Describe two jobs in physical AI. **(4 marks)**

[Total: 20 marks]

Question 15.5: The Final Group Pitch

A student group pitches an AI system to predict malaria drug stock-outs in district health centres.

- a)** Describe the problem and the data the group needs. **(4 marks)**
- b)** Recommend a technique and justify it. **(3 marks)**
- c)** Identify the system's users. **(3 marks)**
- d)** Explain the ethical risks and how to reduce them. **(4 marks)**
- e)** Identify jobs the project could create. **(2 marks)**
- f)** Explain how the project draws on at least four of the course's six review blocks. **(4 marks)**

[Total: 20 marks]